



# A bilevel conic optimization model for routing and charging of EV fleets serving long distance delivery networks

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## ABSTRACT

Recent unveiling of electric semi-trucks by a number of electric vehicle manufacturers indicates that part of the existing long-distance transportation fleets may soon be electrified. Operators of electric fleets will have to select travel routes considering charging station availability and cost of charging in addition to usual factors such as congestion and travel time. This requires combined modeling of transportation and electric power networks. We present such a model that considers interactions between the two networks to develop optimal routing strategies. The problem is formulated as a multi-objective bilevel conic optimization model. The upper level obtains the routing decision by minimizing a function of charging cost and travel time. The routing decision is used in the lower level that solves the AC optimal power flow model, using second order cone constraints, to determine nodal electricity prices. The model is demonstrated using a numerical problem with 24-Node transport network supported by a modified 5-Bus PJM network. The results show that our model yields optimal routes and charging strategies to meet the objectives of fleet operators. Results also indicate that the optimal routing and charging strategies of the electrified transportation fleet can support power networks to reduce nodal prices via demand response.

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## 1. Introduction

The infrastructures supporting electric power and ground transportation will become increasingly co-dependent due to the rapid adoption of electric vehicles (EVs) to promote decarbonization [1,2]. It is projected that by 2023 the number of EVs will grow to 7.3 million [3] in the U.S. and will further increase to 18.7 million by 2030 [4]. The growth of EVs is poised to extend beyond passenger vehicles to semi-trucks supporting long-distance goods transportation. This expected expansion of EV use in the long-range transportation sector makes it imperative to examine both the physical and functional interdependence of electric and transportation networks. This will require a careful analysis of: logistical complexities in long-distance transportation planning, changes in the traffic pattern and charging behavior, impact on electricity network demand patterns, changes in grid ancillary services, and

evolution of economic paradigms guiding power market operations. Two key challenges that arise from the above are 1) route selection of EVs guided by range limitation, charging infrastructure availability, and cost of charging, and 2) assessment of nodal price variations caused by real and reactive power consumption by a large pool of EVs. It may be noted that the nodal prices of electricity will influence the route choice decisions, while in turn, the spatiotemporal distribution of EV charging due to their route choices will influence the nodal prices [5]. In this paper, we develop a model to jointly address the above two challenges by integrating the route choice and charging decisions of EVs and their impact on the electricity network.

We adopt a bilevel optimization framework. The upper level is a transportation model that decides optimal routing for the EV fleets by simultaneously considering the travel time and charging cost. The lower level is an AC optimal power flow (AC-OPF) model that determines the price of electricity at each bus supporting transportation nodes for a given base load and the EV-fleet charging load. The proposed bilevel model is novel as it considers both real

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| Nomenclature                             |                                                                                                                    |                                        |                                                                                                                                       |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sets</b>                              |                                                                                                                    | $V_1$                                  | Nominal voltage at bus 1                                                                                                              |
| $R$                                      | Set of transportation nodes                                                                                        | $F_{kl}$                               | Line flow limit in MW between buses $k$ and $l$                                                                                       |
| $A$                                      | Set of transportation arcs                                                                                         | <b>Variables</b>                       |                                                                                                                                       |
| $T$                                      | Set of discrete time periods, hours                                                                                | $y_{i,t}^v$                            | 1 if fleet $v$ is charging at node $i$ at time $t$ , else 0                                                                           |
| $V$                                      | Set of EV fleets                                                                                                   | $z_{i,t}^v$                            | 1 if fleet $v$ is at node $i$ at time $t$ , else 0                                                                                    |
| $E$                                      | Set of buses in electric network                                                                                   | $p_{i,t}^v$                            | 1 if fleet $v$ arrives at node $i$ at time $t$ , else 0                                                                               |
| $\Omega(k)$                              | Set of electricity line branches connected to bus $k$                                                              | $x_{ij,t}^v$                           | 1 if fleet $v$ departs from node $i$ for node $j$ at time $t$ , else 0                                                                |
| <b>Parameters</b>                        |                                                                                                                    | $s_t^v$                                | State of charge of fleet $v$ at time $t$                                                                                              |
| $\rho$                                   | Relative weight, $0 \leq \rho \leq 1$                                                                              | $u_i$                                  | 1 if a charging station is available at node $i$                                                                                      |
| $\Lambda$                                | Factor to convert travel time into cost (\$/min)                                                                   | $P_{k,t}^g$                            | Real power generation at bus $k$ , at time $t$                                                                                        |
| $\gamma^v$                               | Power dissipation per hour of travel for fleet $v \in V$                                                           | $Q_{k,t}^g$                            | Reactive power generation at bus $k$ , at time $t$                                                                                    |
| $\bar{S}^v, \underline{S}^v$             | Maximum and minimum state of charge for fleet $v$ , respectively                                                   | $R_{kl,t}, I_{kl,t}$                   | Auxiliary decision variables; function of voltage and power angles between buses $k$ and $l$ , at time $t$                            |
| $S_d^v$                                  | Desired state of charge at destination for fleet $v$                                                               | $u_{k,t}$                              | Auxiliary decision variable; function of voltage at bus $k$ , at time $t$                                                             |
| $\bar{P}$                                | Charging rate for fleets                                                                                           | $\alpha_{k,t}$                         | Dual variable corresponding to real power balance constraint at bus $k$ , at time $t$                                                 |
| $C_i$                                    | Number of charging stations at node $i \in R$ , each with capacity to charge whole fleet at any given time         | $\beta_{k,t}$                          | Dual variable corresponding to reactive power balance constraint at bus $k$ , at time $t$                                             |
| $i_0^v, i_d^v$                           | Origin and destination nodes of trip served by fleet $v$                                                           | $\lambda_{kl,t}$                       | Dual variable corresponding to the second-order cone constraint in the power flow formulation between buses $k$ and $l$ , at time $t$ |
| $C_{k,t}$                                | Generation cost function at bus $k$ and at time $t$                                                                | $\eta_{k,t}, \bar{\eta}_{k,t}$         | Dual variables corresponding to lower and upper limits, respectively, of real power generation at bus $k$ , at time $t$               |
| $G_{kl}, B_{kl}$                         | Conductance and Susceptance in the line branches between bus $k$ and $l$ , respectively                            | $\varrho_{kl,t}, \bar{\varphi}_{kl,t}$ | Dual variables corresponding to line flow limits, respectively, between buses $k$ and $l$ , at time $t$                               |
| $P_{k,t}^d, Q_{k,t}^d$                   | Real and reactive power demand, respectively, at bus $k \in E$ and at time $t \in T$                               | $\pi_{k,t}, \bar{\pi}_{k,t}$           | Dual variables corresponding to lower and upper limits, respectively, of reactive power generation at bus $k$ , at time $t$           |
| $\bar{P}_{k,t}^g, \underline{P}_{k,t}^g$ | Maximum and minimum real power generation limits, respectively, in MW at bus $k \in E$ and at time $t \in T$       |                                        |                                                                                                                                       |
| $\bar{Q}_{k,t}^g, \underline{Q}_{k,t}^g$ | Maximum and minimum reactive power generation limits, respectively, in MVAR at bus $k \in E$ and at time $t \in T$ |                                        |                                                                                                                                       |

and reactive power via constraints, including a cone constraint, in determining locational marginal prices (LMPs) of electricity. Existing literature on bilevel formulations for modeling electric power networks use DC-OPF at the lower level and thus do not consider reactive power, for example [6,7]. Additionally, related literature does not consider the direct influence of electricity prices when planning the integration of transport and electric network, as they are normally forecasted or considered to be exogenous to the model [8,9].

Our model formulation is adapted to a medium to long range transportation network served by EV semi-trucks. A limited number of nodes in the transportation network is considered to host charging stations. A large number of travel demands (trips) are considered to be served daily by a number of EV fleets. The trips are characterized by their origin-destination pairs and start times. Each fleet is considered to be comprised of homogeneous vehicle types with same model and battery capacity. However, different fleets may vary in size and type of vehicle used. At the start of a trip, the charge level of each fleet is assumed to be known. The travel network is considered to cover a sufficiently wide region, and thus different subsets of its charging nodes are supported by different power network buses. Hence, the charging stations are subjected to the corresponding nodal prices of electricity, which may vary based on the prevailing base load and the added demand from EVs.

Our work in this paper is motivated by the recent expansions of

the leading EV manufacturers (e.g., Daimler, Nicola, Tesla, and Volvo) towards developing long-distance delivery vehicles (semi-trucks). This is expected to encourage companies with large and expansive distribution networks (e.g., Amazon, Wal-Mart, and UPS in the U.S.) to replace part of their transportation fleets with EVs. Transportation companies with a significant number of fleets with electric semi-trucks will face route choice decisions beyond the existing criteria of congestion, toll rate, and travel time. Availability of charging infrastructure and varying electricity costs at different times of the day and various network locations will influence EV drivers' route choice decisions, thus creating new opportunities for demand response in power networks. Fig. 1 presents a schematic for a distribution network of a large corporation (Wal-Mart) in the State of Florida in U.S [10]. Per information published by Wal-Mart, semi-trucks delivering goods from their distribution centers to their stores across the network travel on an average a distance of 125 miles per trip, and each distribution center services 90-120 stores. Clearly, with increasing long-range vehicle electrification, Wal-Mart alone may generate a significant number of EV trips across the network. Such companies will need to solve their route choice problem by minimizing a function of the travel time and charging cost. Supporting the above need is the key objective of this paper. Our objective also includes demonstrating how the fleet operators' optimal actions will serve as demand response to benefit the power network by load smoothing across zones and hours of



Fig. 1. A hypothetical goods transportation network with supporting power grid in the State of Florida, U.S.A.

the day, which will help to reduce average nodal prices.

## 2. Literature review

The transportation policy makers and urban logistics planners have been leading the ongoing efforts of designing sustainable transportation systems [11] with a focus on improving their environmental, social, and economic impacts. It is anticipated that in near future large fleet operators will reduce their reliance on fossil fuel [12] by adopting electric semi-trucks that are being introduced to the market by auto makers worldwide. The use of EVs by the transportation sector will significantly reduce carbon emissions [13]. The fleet operators will thus face a new task of jointly optimizing their routes and associated charging costs, which would be different from the prevailing approach guided by congestion and toll prices. As the electric semi-trucks may travel hundreds of miles everyday, routing decisions must consider timely delivery of the goods and services as well as minimizing the aggregated charging costs for the fleet operation. This depends on the charging station locations across the travel network. The EV semi-truck charging will also influence the consumption pattern and corresponding price of electricity. The EVs continue to face adoption barrier due to their limited drive range [14], which can be overcome effectively by placing sufficient charging facilities in the transportation network.

There exists a volume of literature that examine optimal location of charging stations. This literature can be grouped into two major categories: one focused on driving range [15,16] and the other on infrastructure cost [17,18]; these cited papers represent only a small subset of the available literature.

Though the current level of penetration of EVs does not significantly affect the electricity grid operations and corresponding pricing patterns, the impending use of a growing number of EV semi-trucks in transportation fleets will surely exert measurable influence on electricity demand, daily load patterns and prices [19]. A recent study of the German grid predict that 42 million new EVs by 2030 will increase daily average demand fluctuations by 92% [20]. According to the study [21], the high expected penetration of EVs will have a noticeable impact on the power grid including need for additional generation capacity and increase in transmission congestion resulting in higher electricity prices. In this context, the interdependence between EV charging and electricity prices is modeled in recent papers [22,23], among others. Authors in Ref. [22] studied the impact of EV charging schemes in power system expansion planning. They found that EVs will promote additional capacity investments in power networks. The authors in Ref. [23] developed a two-layer model for dynamic pricing considering demand response from EV charging stations, which showed that deliberate EV charging strategies can reduce the price

of electricity.

A number of studies are available in the open literature that look into strategic routing decisions of the EVs considering traffic patterns, user equilibrium and exogenous electricity prices without directly considering interdependence with the electricity grid. The authors in Ref. [24] introduced a general electric vehicle routing problem (EVRP) that finds the optimal routing strategy with minimal travel time cost and energy cost while accounting for the EV load effect on energy consumption. Authors in Ref. [25] also considered an EVRP with detailed representation of EV energy consumption and exogenous time-of-use energy prices. Another detailed EVRP model with time varying exogenous prices, different charging modes, and load-speed dependent EV energy consumption can be found in Ref. [26]. Reviews of variants and solution approaches for the EVRP can be found in Refs. [27,28].

In recent years, both researchers and practitioners have noted that the growth of EVs and their interactions with power grids would bring both challenges and opportunities for power networks. Authors in Ref. [29] studied a coupled transportation and power distribution system. They used a best-response approach, which consists of iteratively solving the traffic assignment problem and the optimal power flow problem to find the equilibrium between the two systems. The study in Ref. [8] presents a bi-level charging station planning model to analyze the integration between the power and transportation system. This study, however, do not consider the routing choices of electric vehicles. Similarly, the work presented in Ref. [9] examine the planning of different types of EV charging facilities in an urban area. As in other studies, they also do not considered the EV routing and demand choices, instead use a forecasting framework to model the electricity demand. There are a few papers in the literature that consider impact of electricity loads arriving from EV charging hubs on the operation of the power network [30,31]. These papers use a bi-level modeling approach where the upper level considers energy dispatch by the system operator and the lower level obtains an optimal charging schedule for the EVs. Literature also includes topics on how a mass incorporation of EVs within the electricity grid will be addressed via approaches such as aggregator framework [32], game theoretical models [33,34], and policy options [35].

As evident from the above discussion, there is a gap in the literature concerning the integration of transportation and power networks, which has come to fore by the increasing level of power consumption by EVs. To do so, it is necessary to integrate power dispatch formulations with the existing EVRPs. AC Optimal Power Flow (OPF) formulations are non-convex, solution of which is NP-hard. A simplification approach found commonly in the literature is the use of DC-OPF, where non-linear power flow constraints are linearized and only the real power flow is considered. However, a number of papers in the literature have started to consider AC-OPF, that considers both real and reactive power, with a number of different relaxation methods. Significant among those are convex approximation using Taylor series [36], McCormick envelope [37,38], semi-definite programming (SDP), and second order cone programming (SOCP). Among these, SDP and SOCP have gained more prominence in the recent literature. The SDP relaxation of the OPF problem considers positive semi-definite matrix that ensures convexity of the problem. However for large networks, which are often sparse, SDP relaxation demands both significant memory and more computational time. The SOCP method on the other hand performs well for sparse AC networks as it relaxes the positive semi-definite constraint. Hence, we have adopted the SOCP relaxation approach [39,40] in solving the integrated EVRP and power network flow problem.

In our integrated EVRP and AC-OPF model, we adopted a commonly used EVRP model [41,42]. The scope of this EVRP model

is limited compared to those of [25,26], as it does not considered non-linear battery constraint and speed-load dependent EV energy consumption. Consideration of these would have significantly increased the computational challenges of our integrated model for any reasonably sized transportation network. However, consideration of the power network allows us to model endogenous electricity prices and their effect on the EV route choices. Our integrated model is formulated as a multi-objective bi-level conic optimization problem. The upper level solves the EVRP yielding the route choices and corresponding charging patterns by minimizing a function of travel time and electricity consumption cost. The lower level model considers the AC-OPF with SOCP relaxation that yields the LMPs influenced by the EV routing choices. The multi-objective bi-level program is reformulated as a mathematical program with equilibrium constraints (MPEC) by considering the optimality conditions of the SOCP at the lower level problem. We solved the resulting MPEC model by using a standard state of the art optimization solver.

### 3. The problem and the modeling approach

The problem addressed in this paper is how to optimally route fleets of EV semi-trucks, where each fleet serves an origin-destination (OD) pair from a set of ODs in a long distance transportation network, which is supported by charging stations located at a subset of its nodes. Chargers are assumed to charge at a fixed rate. Different charging nodes are assumed to be supported by different buses of the electric power network. The price of electricity at a bus is determined by the level of power consumption by the EVs as well as the base load that varies with the time of the day. The operating horizon in which all demands are served is considered to be fixed.

The routing decisions for the EV fleets serving a set of OD pairs are guided by two distinct objectives: minimizing the charging cost as well as the travel time. We adopt a bilevel modeling approach (see Fig. 2) where the upper level minimizes a function of the charging cost and the travel time as a multiobjective problem. This decision from the upper level model is fed to the lower-level AC-OPF model, which determines the nodal prices of electricity at the buses (locational marginal prices, LMPs). The AC-OPF is formulated as a second-order cone program (SOCP), for which the appropriate KKT conditions for the conic problem are used to reformulate the conic bilevel problem into a single level multi-objective conic MPEC (mathematical program with equilibrium constraints). The reformulated problem is solved using commonly available mathematical optimization solvers.

In what follows, we present an abstract formulation of the

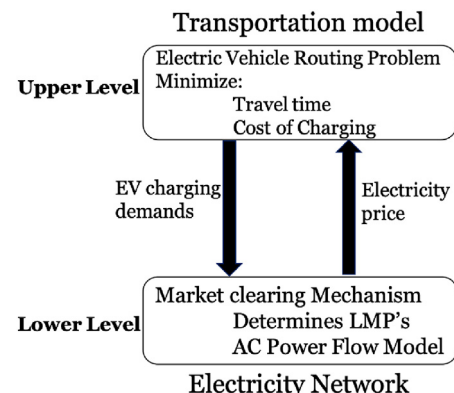


Fig. 2. A bilevel model framework.

multi-objective conic bilevel optimization model.

$$\begin{aligned} \min_x F(x, w) = & [f_1(x, w), f_2(x)] \\ \text{s.t.} & \\ Dx = d, x \in [0, 1] & \\ w \in \arg \min_{y, z} Hz & \\ \text{s.t.} & \\ Ax + By + Cz = b \quad (w) & \\ y \in K \quad (\sigma) & \end{aligned}$$

Two critical features of this conic bilevel model are the multi-objective ( $f_1(\cdot), f_2(\cdot)$ ) nature of the upper-level model and the second-order cone constraint ( $y \in K$ , where  $K$  denotes the cone) of the lower-level model. The upper-level decision variable  $x$  denotes the optimal routing and charging decisions, and  $w$  denotes the price of electricity. The lower-level variable  $y$  is a subset of variables that are used to construct the AC-OPF cone constraints. The variable  $z$  denotes the optimal power-generation schedule. Given within parentheses ( $w$  and  $\sigma$ ) are the dual variables associated with the lower-level constraints.

The single level MPEC reformulation of the above conic bilevel model can be written as follows.

$$\begin{aligned} \min_{x, y, z, w, \sigma} F(x, w) = & [f_1(x, w), f_2(x)] \\ \text{s.t.} & \\ Dx = d, x \in [0, 1] & \quad \text{Upper level constraints} \\ \left. \begin{aligned} Ax + By + Cz = b \\ y \in K \end{aligned} \right\} & \quad \text{Primal feasibility constraints} \\ \left. \begin{aligned} Cw = H \\ Bw + \sigma = 0 \\ \sigma \in K^* \end{aligned} \right\} & \quad \text{Stationarity conditions} \\ w^T(b - Ax) = Hz & \quad \text{Strong duality condition} \end{aligned}$$

In the above single-level MPEC,  $K^*$  is the dual cone of  $K$ . The complementarity condition is met by the strong duality property of the lower-level problem.

#### 4. The model

The problem considered here involves two distinct but interconnected networks: the transportation network at the upper level and the electricity network at the lower level. The transportation network model attempts to minimize the cost as well as the time needed to accomplish the trips, which in turn determines the pattern of power consumption at various buses of the electricity network. This power consumption pattern and the network constraints together dictate the electricity prices at buses, which are fed back to the upper-level transportation model. The models are uniquely different as the transportation model is guided by user preferences, while the electric power model is guided by both the physical laws and market economics.

The parameters and variables used in these models are described in the nomenclature section. It is assumed that the

charging facility locations across the region along the transportation routes are known. Hence, determining the locations of charging stations is not considered in the model.

##### 4.1. Transportation model - upper level problem

The multi-objective function (1) is a  $\rho$  weighted sum of the cost of charging and travel time. The parameter  $\Lambda$  in (1) is a conversion factor (dollars per minute) that converts travel time into cost. Maximizing the total time spent at the destination by all the fleets ( $\sum_{t \in T} \sum_{v \in V} z_{i_d, t}^v$ ) is used as a surrogate for travel time minimization. Let

$R_k$  denotes a subset of transportation nodes with charging stations that are connected to the  $k$ th electricity bus. The LMP of the electricity network corresponds to a dual variable of the lower-level AC-OPF ( $\alpha_{k, t}$ ). Hence, if a EV fleet  $v$  decides to charge at node  $k$  and hour  $t$  ( $y_{i, t}^v = 1$ ), the cost incurred by the fleet is  $\alpha_{k, t} \bar{P}$ .

$$\min \quad \rho \sum_{\substack{t \in T \\ k \in E}} \alpha_{k, t} \left( \sum_{\substack{i \in R_k \\ v \in V}} \bar{P} y_{i, t}^v \right) - (1 - \rho) \Lambda \sum_{\substack{t \in T \\ v \in V}} z_{i_d, t}^v \quad (1)$$

$$\begin{aligned} \text{s.t.} \\ z_{i, t+1}^v = z_{i, t}^v + p_{i, t}^v - \sum_{(i, j) \in A} x_{ij, t}^v \quad \forall i \in R, t \in T, v \in V \end{aligned} \quad (2)$$

$$p_{i, t}^v = \sum_{(j, i) \in A} x_{ji, t}^v \quad \forall i \in R, t \in T, v \in V \quad (3)$$

$$\sum_{i \in R} \left( z_{i, t+1}^v + \sum_{j \in R} x_{ij, t}^v \right) \leq 1 \quad \forall t \in T, v \in V \quad (4)$$

$$z_{i,t}^v + p_{i,t}^v \leq 1 \quad \forall i \in R, t \in T, v \in V \quad (5)$$

$$s_{t+1}^v = s_t^v + \sum_{i \in R} \bar{p}y_{i,t}^v - \gamma^v \left( 1 - \sum_{i \in R} z_{i,t}^v \right) \quad \forall t \in T, v \in V \quad (6)$$

$$\underline{s}^v \leq s_t^v \leq \bar{s}^v \quad \forall t \in \{0, |T| - 1\}, v \in V \quad (7)$$

$$s_{|T|}^v \geq s_d^v \quad \forall v \in V \quad (8)$$

$u_{k,t}$ ,  $P_{k,t}^g$ , and  $R_{kl,t}$ , while the other decision variables  $I_{kl,t}$  and  $Q_{k,t}^g$  are free variables. Some of these variables are defined below as in Ref. [43]:  $u_{k,t} = \frac{V_{k,t}^2}{\sqrt{2}}$ ,  $R_{kl,t} = V_{k,t}V_{l,t}\cos\theta_{kl,t}$ ,  $I_{kl,t} = V_{k,t}V_{l,t}\sin\theta_{kl,t}$ , where  $V_{\cdot,t}$  and  $\theta_{kl,t}$  are the bus voltage magnitude and phase angle between buses, respectively.

$$\min \sum_{t \in T} \sum_{k \in E} c_{k,t} P_{k,t}^g \quad (11)$$

$$P_{k,t}^g - \sqrt{2}u_{k,t} \sum_{l \in \Omega(k)} G_{kl} + \sum_{l \in \Omega(k)} (G_{kl}R_{kl,t} - B_{kl}I_{kl,t}) = P_{k,t}^d + \sum_{i \in R} \sum_{v \in V} \bar{p}y_{i,t}^v \quad \forall t \in T, \forall k \in E \quad (\alpha_{k,t}) \quad (12)$$

$$Q_{k,t}^g - \sqrt{2}u_{k,t} \sum_{l \in \Omega(k)} B_{kl} + \sum_{l \in \Omega(k)} (B_{kl}R_{kl,t} + G_{kl}I_{kl,t}) = Q_{k,t}^d + \omega \sum_{i \in R} \sum_{v \in V} \bar{p}y_{i,t}^v \quad \forall t \in T, \forall k \in E \quad (\beta_{k,t}) \quad (13)$$

$$y_{i,t}^v \leq z_{i,t}^v \quad \forall t \in T, v \in V, \\ i \in R \setminus \{i_0^v, i_d^v\} \quad (9)$$

$$\sum_{v \in V} y_{i,t}^v \leq C_i u_i \quad \forall t \in T, i \in R \quad (10)$$

Constraint (2) identifies if the location of fleet  $v$  is at a node  $i$  at time  $t + 1$ . The fleet is at node  $i$  if 1) the fleet was already idling at node  $i$  and continued till time  $t + 1$  and 2) the fleet arrived at node  $i$  at time  $t$  and decided to idle till time  $t + 1$ . The fleet  $v$  is not at node  $i$  if it has departed for node  $j$  at time  $t$ . Constraint (3) determines if fleet  $v$  arrives at node  $i$  at time  $t$ . Constraint (4) ensures that a fleet can either stay at a node  $i$  from time  $t$  to  $t + 1$ , or depart from it. Constraint (5) ensures that at any time  $t$ , a fleet can either arrive at node  $i$ , or is already idling at that node, or neither. State of charge for each EV fleet at time  $(t + 1)$  is updated through constraint (6); it considers the state of charge at  $t$ , any additional charge received between  $t$  and  $t + 1$ , and any charge used at the rate  $\gamma^v$  due to travel. The state of charge of a fleet is bounded as given by (7). The desired state of charge at destination is maintained by constraint (8). A fleet can charge at node  $i$  only if it is at that node, see constraint (9). Constraint (10) limits the number of fleets that can charge simultaneously at a given node. The EVRP described above is similar those presented in Refs. [41,42]. Note that other variants of the model can be considered, such as partial recharging, (non) linear battery and consumption constraints, charging modes, among others. For variants of the EVRP refer to Refs. [27,28].

#### 4.2. Power network model - lower level problem

This model adopts a SOCP relaxation of AC-OPF formulation as presented in (11) - (20). We do not consider a separate cost for reactive power, which is assumed to be supplied by the generators. The dual variable  $\alpha_{k,t}$  (\$/MWh) corresponding to the real power balance constraint (12) represents the LMP of electricity. This also accommodates loss due to reactive current flow. Since the reactive power cost is assumed to be zero, the dual variable  $\beta_{k,t}$  (\$/MVarh) corresponding to the reactive power balance constraint (13) is negligible. The non-negative decision variables in the model are

$$\begin{pmatrix} R_{kl,t} \\ I_{kl,t} \\ u_{k,t} \\ u_{l,t} \\ u_{k,t} + u_{l,t} \end{pmatrix} \in K \subseteq \mathbb{R}^5 \quad \forall t \in T, l \in \Omega(k), k \in E \quad (\lambda_{kl,t}^{1,\dots,5}) \quad (14)$$

$$R_{kl,t} = R_{lk,t} \quad \forall t \in T, l \in \Omega(k), k \in E \quad (\psi_{kl,t}) \quad (15)$$

$$I_{kl,t} = -I_{lk,t} \quad \forall t \in T, l \in \Omega(k), k \in E \quad (\chi_{kl,t}) \quad (16)$$

$$u_{1,t} = \frac{V_1^2}{\sqrt{2}} \quad \forall t \in T \quad (\mu_t) \quad (17)$$

$$\underline{P}_{k,t}^g \leq P_{k,t}^g \leq \bar{P}_{k,t}^g \quad \forall t \in T, k \in E \quad (\eta_{k,t}, \bar{\eta}_{k,t}) \quad (18)$$

$$\underline{Q}_{k,t}^g \leq Q_{k,t}^g \leq \bar{Q}_{k,t}^g \quad \forall t \in T, k \in E \quad (\pi_{k,t}, \bar{\pi}_{k,t}) \quad (19)$$

$$F_{kl} \geq -\sqrt{2}u_{k,t}G_{kl} + G_{kl}R_{kl,t} - B_{kl}I_{kl,t} \geq -F_{kl} \\ \forall t \in T, l \in \Omega(k), k \in E \quad (\varphi_{kl,t}, \varphi_{kl,t}) \quad (20)$$

Constraints (12) and (13) represent real and reactive power balance of the network considering base active power load (residential and business)  $P_{k,t}^d$  MW, reactive power load  $Q_{k,t}^d$  MVar, and the power consumed by the fleets;  $\omega$  denotes the fraction of real power consumed as reactive power by the fleets. Presented in (14) is the second-order cone constraint of the AC-OPF model, which is reformulated from  $2u_{k,t}u_{l,t} = R_{kl,t}^2 + I_{kl,t}^2$  (see Ref. [43]). Additional equations required to solve for the decision variables are given in (15) and (16). The voltage at reference bus (or slack bus) is known and is given in (17). The limits of real and reactive power generation, and the line flow are considered in (18), (19) and (20), respectively. The dual variable(s) of each constraint are provided

within parentheses next to each constraint.

#### 4.3. Single level model

The bilevel model is reformulated into a single-level (conic) mixed integer programming model using the corresponding KKT conditions for conic programs, which include the set of primal constraints (12) – (20), collection of stationarity conditions with respect to dual variables (22) – (27), and considering the dual cone property (28). These conditions are valid as the conic constraints are continuous and convex. The strong duality constraint (29) is imposed, as SOCP problem may lead to non-zero gap solution, on primal and dual objective functions to ensure optimality.

$$\min \quad \rho \sum_{\substack{t \in T \\ k \in E}} \alpha_{k,t} \left( \sum_{\substack{i \in R_k \\ v \in V}} \bar{P} y_{i,t}^v \right) - (1 - \rho) \Lambda \sum_{\substack{t \in T \\ v \in V}} z_{i_d,t}^v \quad (21)$$

s.t.

Constraint (2) – (10), (12) – (20)

$$G_{kl}\alpha_{k,t} + B_{kl}\beta_{k,t} + \lambda_{kl,t}^1 + \psi_{kl,t} - \psi_{lk,t} + G_{kl}\varphi_{kl,t} - G_{kl}\bar{\varphi}_{kl,t} \leq 0 \quad l \in \Omega(k), k \in E, \forall t \in T \quad (R_{kl,t}) \quad (22)$$

$$-B_{kl}\alpha_{k,t} + G_{kl}\beta_{k,t} + \lambda_{kl,t}^2 + \chi_{kl,t} + \chi_{lk,t} - B_{kl}\varphi_{kl,t} + B_{kl}\bar{\varphi}_{kl,t} = 0 \quad l \in \Omega(k), k \in E, \forall t \in T \quad (I_{kl,t}) \quad (23)$$

$$\left. \begin{aligned} & -\sqrt{2} \alpha_{k,t} \sum_{l \in \Omega(k)} G_{kl} - \sqrt{2} \beta_{k,t} \sum_{l \in \Omega(k)} B_{kl} + \sqrt{2} \sum_{l \in \Omega(k)} G_{kl}\bar{\varphi}_{kl,t} \\ & -\sqrt{2} \sum_{l \in \Omega(k)} G_{kl}\varphi_{kl,t} + \sum_{l \in \Omega(k)} (\lambda_{kl,t}^3 + \lambda_{kl,t}^5) + \sum_{l \in \Omega(k)} (\lambda_{lk,t}^4 + \lambda_{lk,t}^5) \end{aligned} \right\} \leq 0 \quad \forall t \in T, k \in E \setminus \{1\} \quad (u_{k,t}) \quad (24)$$

$$\left. \begin{aligned} & -\sqrt{2} \alpha_{1,t} \sum_{l \in \Omega(1)} G_{1l} - \sqrt{2} \beta_{1,t} \sum_{l \in \Omega(1)} B_{1l} + \sqrt{2} \sum_{l \in \Omega(1)} G_{1l}\bar{\varphi}_{1l,t} \\ & -\sqrt{2} \sum_{l \in \Omega(1)} G_{1l}\varphi_{1l,t} + \sum_{l \in \Omega(1)} (\lambda_{1l,t}^3 + \lambda_{1l,t}^5) + \sum_{l \in \Omega(1)} (\lambda_{l1,t}^4 + \lambda_{l1,t}^5) \end{aligned} \right\} \leq -\mu_t \quad \forall t \in T \quad (u_{1,t}) \quad (25)$$

$$\begin{aligned} \sum_{k \in E} c_{k,t} P_{k,t}^g = & \sum_{k \in E} \left[ \left( P_{k,t}^d + \sum_{i \in R_k} \sum_{v \in V} \bar{P} y_{i,t}^v \right) \alpha_{k,t} + \left( Q_{k,t}^d + \omega \sum_{i \in R_k} \sum_{v \in V} \bar{P} y_{i,t}^v \right) \beta_{k,t} - \bar{P}_{k,t}^g \bar{\eta}_{k,t} \right. \\ & \left. + P_{k,t}^g \eta_{k,t} - \bar{Q}_{k,t}^g \bar{\pi}_{k,t} + Q_{k,t}^g \pi_{k,t} + \sum_{l \in \Omega(k)} (-F_{kl}\varphi_{kl,t} - F_{kl}\bar{\varphi}_{kl,t}) \right] + \frac{V_1^2}{\sqrt{2}} \mu_t \quad \forall t \in T \end{aligned} \quad (29)$$

Note that, the objective function (21) has a non-linear term  $y_{i,t}^v \alpha_{k,t}$  and the constraint (29) has non-linear components  $y_{i,t}^v \alpha_{k,t}$  and  $y_{i,t}^v \beta_{k,t}$ . These non-linear terms can be linearized using big M method. Let  $Z_{k,i,t}^v$  be a new decision variable and the nonlinear term  $y_{i,t}^v \alpha_{k,t}$  can then be linearized as:  $Z_{k,i,t}^v \leq M y_{i,t}^v$ ,  $Z_{k,i,t}^v \geq \alpha_{k,t} - M(1 - y_{i,t}^v)$ , and  $Z_{k,i,t}^v \geq 0$ . The big M takes the maximum value of  $c_{k,t}$ . A similar approach is adopted to linearize  $y_{i,t}^v \beta_{k,t}$ .

## 5. Model parameters and numerical results

We construct a sample application problem by using a transportation network supported by an electricity network, as presented in Fig. 1, to demonstrate implementation of our model. The transportation network is the 24-Node Sioux Falls network (see Ref. [44]) and the supporting electricity network is a modified PJM 5-Bus network. As shown in Fig. 1, each of the five electricity buses (A, B, C, D, and E) are connected to the surrounding transportation nodes. The electricity network incorporates line congestion resulting in five different LMPs and corresponding charging cost at the buses. We consider 12 h (7AM-7PM) of daily operation for the transportation network.

In the following subsection, we provide the model parameter values including those for 5-Bus PJM and 24-Node Sioux Falls networks and the electric vehicles. We apply our model on the sample application problem to obtain 1) the Pareto front for routing decision of the fleet operators guided by different objectives, and 2) the demand response action offered by EVs. The model is implemented using the Julia language and solved with GUROBI-8 in a HPE DL360 Gen10 Xeon server with 64 GB of RAM. We limited the maximum run time for each instance of the model to 8 h.

### 5.1. Experimental setting and model parameters

The parameter  $\rho \in [0, 1]$  used in the objective function accounts for the fleet operator's preference between minimizing EV charging cost and travel time. Also, the conversion parameter  $\Lambda$  in the objective function is assumed to be \$ 0.20/min, as considered in Ref. [45]. The modified PJM 5-Bus network has four power generators located at nodes A, C, D, and E. The generator capacities are 400, 520, 400, and 600 MW, respectively. The generator production costs are 13, 18, 25, and 30 \$/MWh, respectively. The base active power consumption (baseload) by the residential, commercial, and industrial loads (other than EVs) are varied between 200 MW and 1300 MW during the hours of a day. The base reactive power load is assumed to be 25% of the active load and supported by the generators. The baseload is considered to reach peak at 3 p.m. and valley at 7 PM.

We considered seven charging stations in the transportation network. The location of the seven charging stations and the supporting electricity bus numbers (within parentheses) are as follows: 2 (A), 6 (B), 10 (B), 14 (C), 17 (D), 20 (E), 24 (C). The charging station locations are chosen such that they are distributed across the transportation network. For modeling simplicity, the transit times between all adjacent transportation nodes are assumed to be same, an hour. We consider 50 origin-destination (OD) combinations, where each OD is served by a fleet of 100 identical EV semi-trucks. Having an identical fleet of vehicles is a common practice among transportation companies. The battery capacity of each vehicle is assumed to be 500 kWh. It is also assumed that the batteries maintain a minimum charge level of 20 kWh and the desired minimum remaining charge level upon reaching a destination node is 30 kWh. Energy consumed by the EVs while traveling is 20 kW/h and the maximum charging rate at all stations is

40 kW/h.

### 5.2. Results

The bilevel conic optimization model is solved for nine different scenarios with  $\rho = 0.1, 0.2, \dots, 0.9$ , where  $\rho = 0.1$  corresponds to the scenario where focus is mostly on travel time minimization rather than charging cost, and vice versa when  $\rho = 0.9$ . The model is solved to optimality (with optimality GAP < 0.01%) for seven of the nine scenarios. The two cases that did not solve to optimality achieved a GAP of 6% (for  $\rho = 0.7$ ) and 9% (for  $\rho = 0.8$ ) after reaching a maximum allocated solution time of 8 h. The solutions obtained from these scenarios generate the Pareto envelope showed on Fig. 3. Such an envelope represents the trade-off between travel time and charging cost. In Fig. 3 the 0% point on the x-axis represents the lowest travel time ( $\rho = 0.1$ ) to serve all fifty OD pairs and highest cost of charging. Similarly, the 40+% point represents the scenario with highest travel time and the lowest charging costs ( $\rho = 0.9$ ). It may be noted, minimum travel time (0%) corresponds to a 17% increase in the cost of charging. The dots on the Pareto front for  $\rho$  values 0.2 and 0.4 overlaps with the other points and hence are not labeled.

Clearly, for different values of  $\rho$ , the routing, as well as the charging patterns of the fleets, are expected to be different. For charging cost minimization (with higher  $\rho$  values), the vehicles are likely to take longer routes through areas of lower network base-load with lower nodal prices of electricity, thus incurring reduced cost of charging. Whereas, for travel time minimization (with lower  $\rho$  values), the fleets are likely to adopt more direct routes through areas with likely higher nodal electricity prices. Since prices vary across the hours of the day, irrespective of the chosen route, the fleets optimize their consumption pattern to reduce cost. Fig. 4 presents unique consumption patterns for all values of  $\rho$  over all hours of operation. For different values of  $\rho$ , routing variations are also evident from their charging patterns. It can be seen that for  $\rho = 0.1$ , a significant portion of charging is concentrated in early hours of operation (hour 5) at nodes 10, 14, and 20. Whereas for  $\rho = 0.8$  and 0.9, charging of nodes 10, 14 and 20 is partially shifted to late hours of operation (two peaks) when the nodal prices are assumed to be lower. We have also noted from our results that, since the electricity consumption of the EVs plays a significant role in affecting the nodal prices, shifting the demand of EVs (as in the two peak scenarios) also helps to keep the nodal prices lower. It can also be seen from the Fig. 4 (See y-axis on the left) that EVs

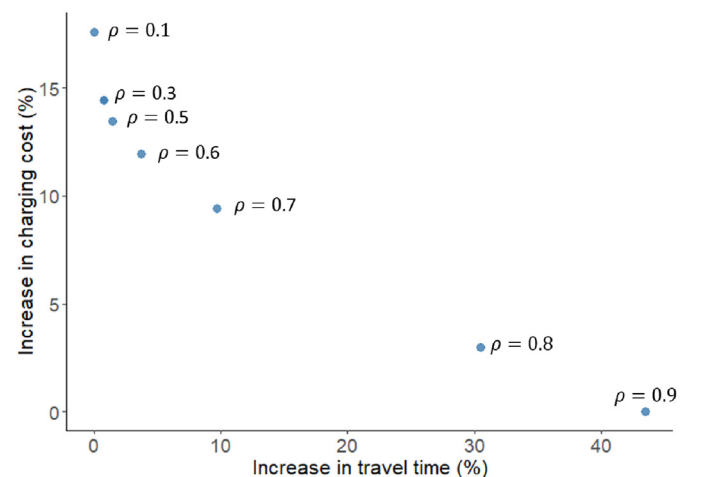


Fig. 3. Pareto curve.

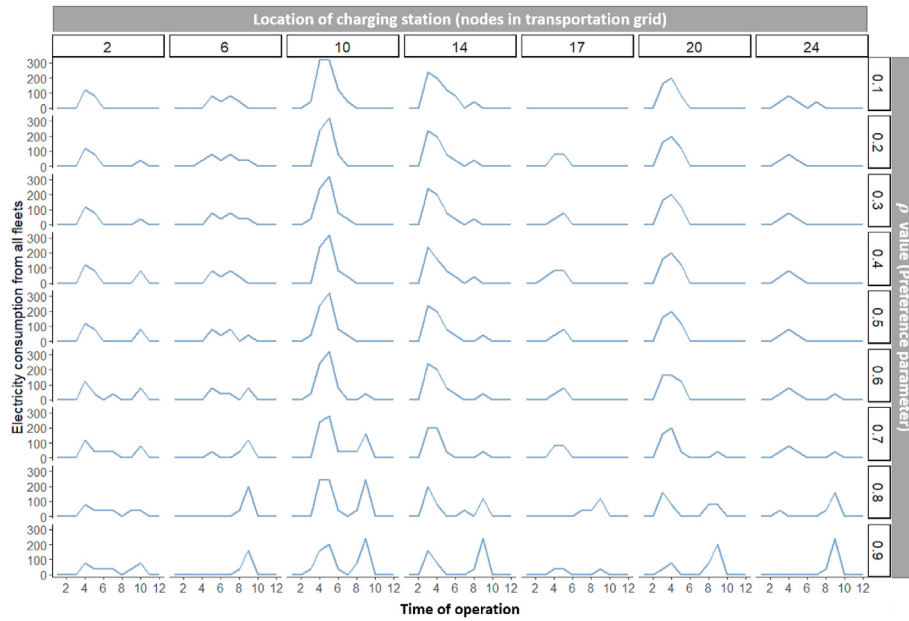


Fig. 4. Demand by charging node and driver preference.

consume minimal to no charge at around 3 p.m. (7th and 8th hours of operation) to avoid contributing to price peaking since the baseload at that time is already high. There is also a noticeable change in the routing and charging patterns as  $\rho$  increases from 0.1 to 0.9. At high values of  $\rho$  the power network operations benefit from load smoothing and corresponding nodal price reduction.

In Fig. 5, the bar plot depicts the distribution of power consumption by the EVs across all operating hours, for two extreme values of  $\rho$  (0.1 and 0.9). At  $\rho = 0.1$ , the EV fleets' primary objective minimize the travel time, in which case the fleet vehicles are charging more at early hours of operation (hours 3–5) even though the base loads at those hours are generally higher (see inset curve in Fig. 5 for baseload distribution). This yields a higher PAR (peak-to-average ratio) of hourly demand and results in a daily average LMP of \$ 23.7/MWh. In contrast, for  $\rho = 0.9$ , power consumption for charging is shifted to later hours of operations when the baseload is

lower. This yields a relatively lower PAR and a lower daily average LMP of \$ 22.3/MWh. This lowering of LMP is attributed to the demand response that results from the fleet's aim to minimize the charging cost. This impact on the hourly cost of electricity (hourly LMPs) resulting from the two modes of fleet operations ( $\rho = 0.1$  and 0.9) is depicted by the line graphs in Fig. 5, which show the deviations of the hourly LMPs from the daily average LMP. Clearly,  $\rho = 0.9$  yields a lower deviation of hourly LMPs.

Fig. 6 exhibits the patterns of power consumption by the fleet for several values of  $\rho$  between 0.1 and 0.9. Clearly, as  $\rho$  increases (i.e., the cost of charging is emphasized more), the charging pattern shifts to later hours of operation. The inset table in the figure shows the cumulative number of fleets reaching their destinations at different hours of operation between 2 and 12. It may be noted again that for lower values of  $\rho$ , more fleets (out of a total of 50) reach their destinations earlier compared to those for higher values

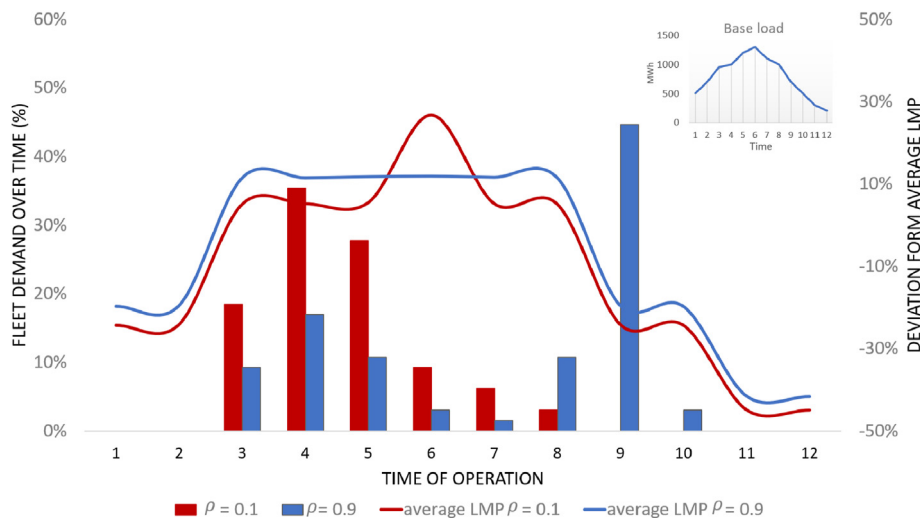
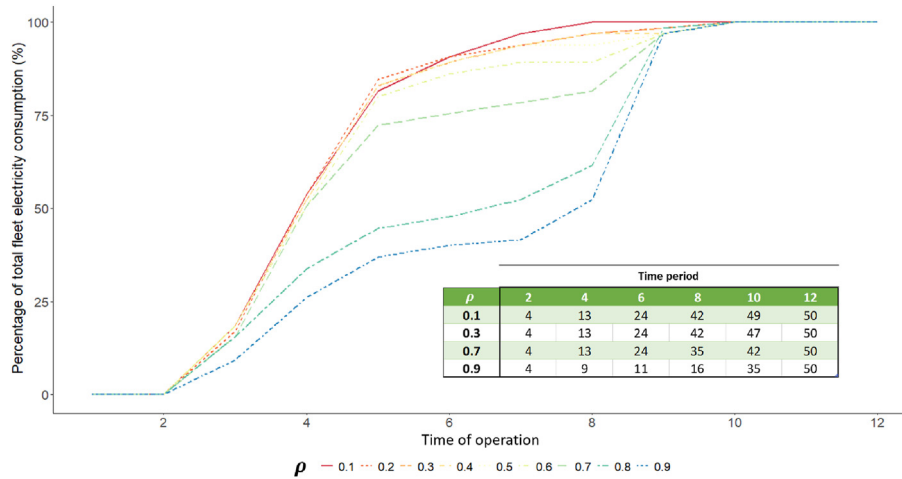


Fig. 5. Distribution of EV charging consumption across the operating hours, for scenarios with  $\rho = 0.1$  and 0.9, and their impact on the deviation of hourly prices from the daily average LMP.



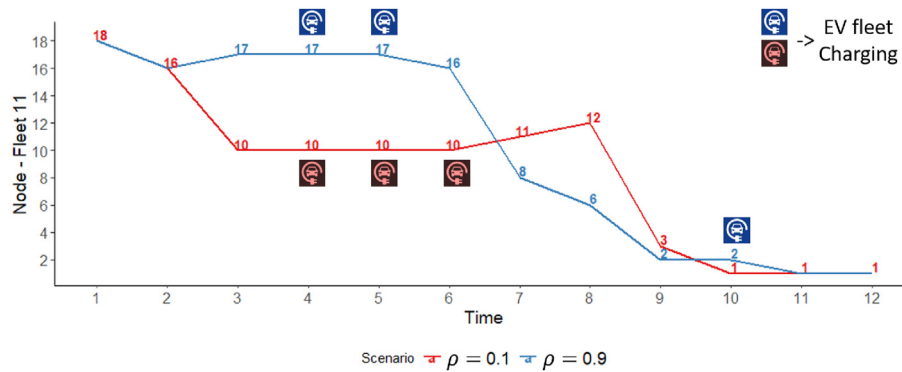
**Fig. 6.** Cumulative distribution of power consumption (line graphs) and the time to reach destinations for the fleets (inset table), for different values of  $\rho$

of  $\rho$ .

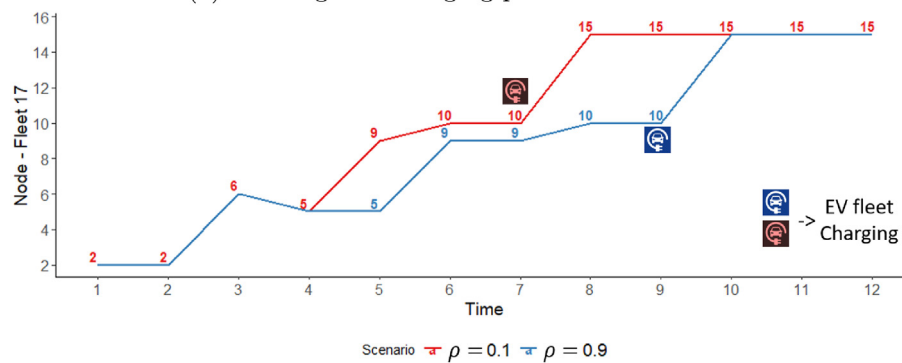
Fig. 7 shows details of the route choices and charging pattern for two distinctively different fleets, number 11 and 17 in our numerical problem. Origin and destination for fleet 11 are nodes 18 and 1, which for fleet 17 are 2 and 15, respectively. As shown in a Fig. 7a, fleet 11 adopts two distinctly different routes for  $\rho = 0.1$  (red) and  $\rho = 0.9$  (blue); see the node numbers on the lines. For  $\rho = 0.1$ , charging takes place at node 10 at h 4, 5 and 6. Whereas for  $\rho = 0.9$ ,

though some charging do take place at hours 4 and 5 at node 17, charging at hour 6 is shifted to hour 10. Note that cost of charging at hour 6 is \$ 30/MWh while at hour 10 is \$ 17/MWh.

In Fig. 7b, we observe an interesting pattern for fleet 17, where for both  $\rho = 0.1$  and 0.9, the fleet adopts the same route and also charges only once. Though in both cases charging occurs at node 10, for  $\rho = 0.9$ , the fleet decides to idle at nodes 5, 9, and 10 to benefit from lower cost of charging at hour 9. The cost of charging at node



(a) Routing and charging patterns for fleet 11



(b) Routing and charging patterns for fleet 17

**Fig. 7.** Routing and charging patterns for the two representative fleets for two extreme values of  $\rho$

10 is \$25/MWh at hour 7 and \$18/MWh at hour 9.

## 6. Concluding remarks

The EV industry has recently expanded its focus to electrifying long-range semi-trucks. Several EV manufacturers have already unveiled their semi-trucks prototypes. Hence it is expected that soon a significant fraction of the long-distance transportation fleets may be EVs. Consequently, the fleet managers will have to route their fleet vehicles considering the spatio-temporal variations in the cost of electricity across the transportation network nodes and the hours of the day. For example, a fleet can be routed through areas where the baseload of electricity (and hence the LMP) is low to reduce the cost of charging while perhaps increasing the travel time. It may also be noted that, due to the interdependence of routing and the power network configuration, if more fleets are routed through a particular node for charging, it will increase the LMP of that node due to higher consumption from charging. Hence, new modeling approaches are needed to consider interaction between EV transportation and electric power networks. This paper presents an approach that is formulated as a multi-objective conic bilevel optimization model. The multi-objective component considers the trade-off between the travel time and charging cost. The second-order cone constraint is used to accommodate the AC optimal power flow model. The upper level of the bilevel model obtains routing decisions, and the lower-level solves the AC optimal power flow model to obtain the nodal prices of electricity resulting from routing decisions.

The model is implemented on a sample problem constructed using a standard transportation network (24-Node Sioux falls) supported by a modified PJM 5-Bus electricity network. The sample problem considers routing of 50 fleets (each with 100 identical semi-trucks) serving various origin and destination combinations. The results show that for different objectives of the fleet operator, our model yields distinctly different routes and charging strategies that are optimal. Our results also demonstrate how optimal routing and charging of the fleets generate demand response and thus support power network to reduce nodal prices. Our model obtains solutions for most instances of the numerical problem with a optimality gap of 0.01%. For networks with a large number of nodes served by a huge number of fleets, the model run time and also the optimality gap will likely be higher. We did not yet implement our model on larger networks to assess computational complexity and solution quality. We believe that the model implementation can benefit from distributed computing that can reduce the computation challenges for problems with large networks. Also, even with some optimality gap, a solution from our model will offer significant decision support for transportation companies willing to transform their fleets with EVs. Hence, the consequences of our model will also be manifested via faster electrification of the transportation fleets and a reduction of vehicular carbon emissions.

## Author statement

All authors reviewed and approved the final version of the manuscript. Vignesh Subramanian and Felipe Feijoo: Conceptualization, methodology, programming, writing, and supervision. Sriram Sankaranarayanan, and Kevin Melendez: Writing, conceptualization, and methodology. Tapas K. Das: Conceptualization and writing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have

appeared to influence the work reported in this paper.

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